

Semester	T.E. Semester VI - CMPN
Subject	DevOps Lab
Subject Professor In-charge	Dr. Snehal Andhare
Assisting Teachers	Dr. Snehal Andhare

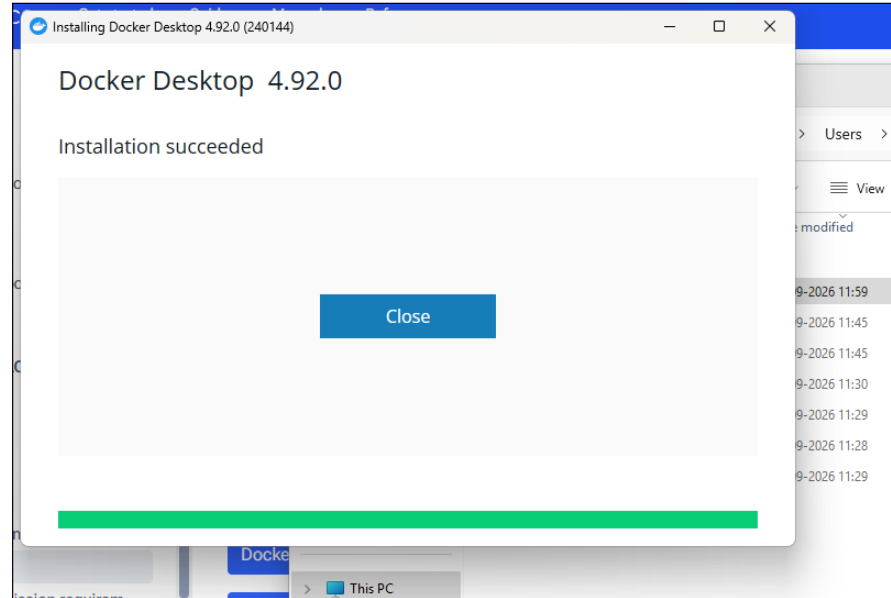
Laboratory

Student Name	Vijay Hanumandla		
Roll Number	23102C0071		
Grade and Subject Teacher's Signature			

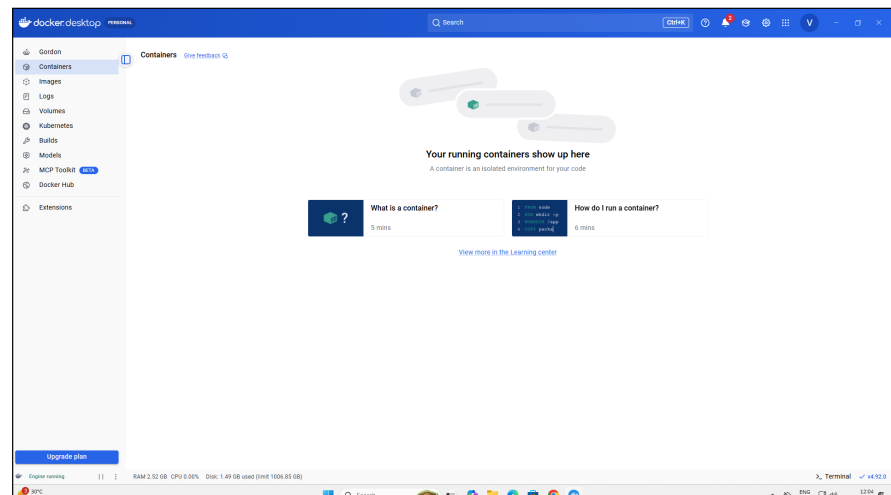
Experiment Number	08
Experiment Title	To learn Dockerfile instructions, build an image for a sample web application using Docker Hub and Docker Desktop.
Objectives (Skill Set / Knowledge Tested / Imparted)	The objective is to learn Dockerfile instructions, build an image for a sample web application, run it as a container, verify the environment variable and copied web files, exercise container lifecycle operations, and publish/retrieve the image using Docker Hub.
Description	This experiment starts with Docker Desktop and Docker Engine verification, then creates an index.html web application and a Dockerfile based on nginx:alpine . The image is built as mywebapp:v1 , run on localhost:8080 , inspected with Docker commands and an interactive shell, stopped and restarted, then tagged, pushed to Docker Hub, removed locally, pulled again, and rerun.
Program	Dockerfile Instructions, Image Build, Container Lifecycle, and Docker Hub Workflow Key commands demonstrated: docker --version; docker info docker build -t mywebapp:v1 . docker images; docker run -d -p 8080:80 --name docker-web-container mywebapp:v1 docker ps; docker exec ...; docker stop/start docker login; docker tag; docker push; docker rm; docker rmi; docker pull

Environment Setup

Task 1: Docker Desktop Installation



Task 2: Start and Verify Docker Desktop



Project & Dockerfile

Task 3: Create Project Folder

```
PS C:\WINDOWS\system32> mkdir C:\DockerLab

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----          24-09-2026     12:05             DockerLab

PS C:\WINDOWS\system32> cd C:\DockerLab
PS C:\DockerLab> mkdir docker-web-demo

Directory: C:\DockerLab

Mode                LastWriteTime         Length Name
----                -
d-----          24-09-2026     12:05             docker-web-demo

PS C:\DockerLab> cd docker-web-demo
PS C:\DockerLab\docker-web-demo> dir
PS C:\DockerLab\docker-web-demo> |
```

Task 4: Create index.html

```
PS C:\DockerLab\docker-web-demo> notepad index.html
PS C:\DockerLab\docker-web-demo> dir

Directory: C:\DockerLab\docker-web-demo

Mode                LastWriteTime         Length Name
----                -
-a-----          24-09-2026     12:07             344 index.html

PS C:\DockerLab\docker-web-demo> |
```

Task 5: Create Dockerfile

```
PS C:\DockerLab\docker-web-demo> Rename-Item "Dockerfile.txt" "Dockerfile"
PS C:\DockerLab\docker-web-demo> dir

Directory: C:\DockerLab\docker-web-demo

Mode                LastWriteTime         Length Name
----                -
-a-----          24-09-2026     12:08             180 Dockerfile
-a-----          24-09-2026     12:07             344 index.html

PS C:\DockerLab\docker-web-demo> |
```

Image Build & Web Access

Task 6: Build Docker Image

```
PS C:\DockerLab\docker-web-demo> docker build -t mywebapp:v1 .
[*] Building 9.1s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 219B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/nginx:alpine@sha256:1ed1b0c1d7652977d6c8da4f018c7b5fc089a7ddc38077351e2eddda745de43f
=> resolve docker.io/library/nginx:alpine@sha256:1ed1b0c1d7652977d6c8da4f018c7b5fc089a7ddc38077351e2eddda745de43f
=> sha256:e72112c14215406dc7f23ef8e5cfddeb8662176167382ad64ec86f52b7a88e 20.53MB / 20.53MB
=> sha256:e72112c14215406dc7f23ef8e5cfddeb8662176167382ad64ec86f52b7a88e 1.40kB / 1.40kB
=> sha256:64c819408f9e2c83e81cc561c5542733627d2c7de493a828139677025f975c 1.21kB / 1.21kB
=> sha256:9a2a404fdd6a3681d43dcce0c0cf374efc702af63311379ea83692498a8 40kB / 40kB
=> sha256:745dfb2698dd8627e1593941cacc756f10943df7938a478bebe4e88c5695cb9 965B / 965B
=> sha256:6c53d0b2a666ed8f8fad8d9982e38cf24eb0885036c17b27591665ca25e1ba40c 629B / 629B
=> sha256:d9aae54b58315e84b9c3ac5f0b0ad6610e0e28ad9ff28fcd694647e0be13131 1.93MB / 1.93MB
=> sha256:e3de96513ba0b53b4b1787ec8a65cddc388ac4772a3e4cd4714dcfde2a102b5 3.85MB / 3.85MB
=> extracting sha256:e2de96513ba0b53b4b1787ec8a65cddc388ac4772a3e4cd4714dcfde2a102b5
=> extracting sha256:d9aae54b58315e84b9c3ac5f0b0ad6610e0e28ad9ff28fcd694647e0be13131
=> extracting sha256:6c53d0b2a666ed8f8fad8d9982e38cf24eb0885036c17b27591665ca25e1ba40c
=> extracting sha256:745dfb2698dd8627e1593941cacc756f10943df7938a478bebe4e88c5695cb9
=> extracting sha256:9a2a404fdd6a3681d43dcce0c0cf374efc702af63311379ea83692498a8
=> extracting sha256:64c819408f9e2c83e81cc561c5542733627d2c7de493a828139677025f975c
=> extracting sha256:e72112c14215406dc7f23ef8e5cfddeb8662176167382ad64ec86f52b7a88e
=> extracting sha256:e72112c14215406dc7f23ef8e5cfddeb8662176167382ad64ec86f52b7a88e
=> [internal] load build context
=> => transferring context: 283B
=> [2/4] WORKDIR /usr/share/nginx/html
=> [3/4] RUN rm -f index.html
=> [4/4] COPY index.html
=> exporting to image
=> exporting layers
=> exporting manifest sha256:8870f97e6ff4143af72de5b08a55d136769854f3ff417869deaa9bb211bf214b9
=> exporting config sha256:84439fd7c0e61b01997f199ed0bc1cd980e14941e3a492ee461fa62a5906cf9
=> exporting attestation manifest sha256:8e11492da17a24dc2220fa91ec15f8a3491323daa6f7ea8d924864ccfd16796
=> exporting manifest list sha256:ed186ce1a49281e0c28881a73f153484e83c58bb0bcedccf8397e3db8abaf
=> naming to docker.io/library/mywebapp:v1
=> unpacking to docker.io/library/mywebapp:v1
PS C:\DockerLab\docker-web-demo>
```

Task 7: View Docker Images

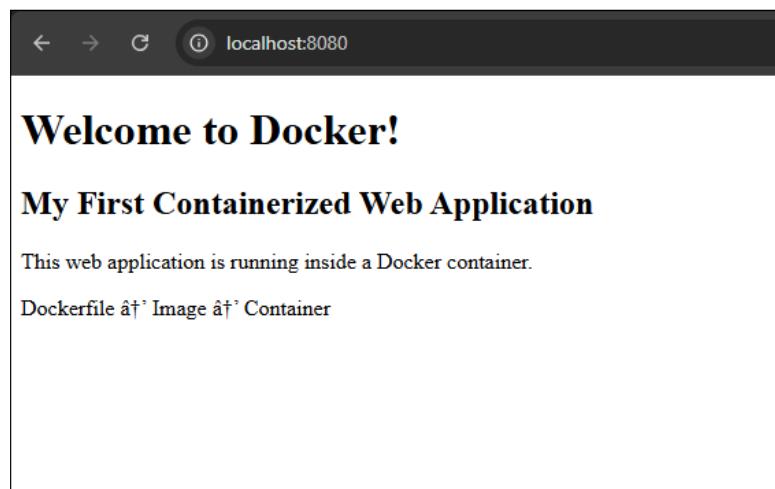
```
PS C:\DockerLab\docker-web-demo> docker images

IMAGE          ID                                DISK USAGE  CONTENT SIZE  EXTRA
mywebapp:v1    ed106ce1a492                    93.6MB      26.3MB
PS C:\DockerLab\docker-web-demo>
```

Task 8: Run Container

```
PS C:\DockerLab\docker-web-demo> docker run -d -p 8080:80 --name docker-web-container mywebapp:v1
adc9196ce9fb4553ded69be09c8e1ccf1401c615f64ab96ac4b48fd3d26db069
PS C:\DockerLab\docker-web-demo>
```

Task 9: Open Web Page



Container Inspection

Task 10: View Running Container

```
PS C:\DockerLab\docker-web-demo> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
ad9196ce8fb   mywebapp:v1   "/docker-entrypoint..." 2 minutes ago  Up 2 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  docker-web-container
```

Task 11: Verify APP_VERSION

```
PS C:\DockerLab\docker-web-demo> docker exec docker-web-container printenv APP_VERSION
1.0
PS C:\DockerLab\docker-web-demo> |
```

Task 12: Enter Container Shell

```
PS C:\DockerLab\docker-web-demo> docker exec docker-web-container printenv APP_VERSION
1.0
PS C:\DockerLab\docker-web-demo> docker exec -it docker-web-container sh
/usr/share/nginx/html # |
```

Task 13: Check Working Directory

```
PS C:\DockerLab\docker-web-demo> docker exec docker-web-container printenv APP_VERSION
1.0
PS C:\DockerLab\docker-web-demo> docker exec -it docker-web-container sh
/usr/share/nginx/html # pwd
/usr/share/nginx/html
/usr/share/nginx/html # |
```

Task 14: List Container Files

```
PS C:\DockerLab\docker-web-demo> docker exec -it
/usr/share/nginx/html # pwd
/usr/share/nginx/html
/usr/share/nginx/html # ls
50x.html    index.html
/usr/share/nginx/html # |
```

Task 15: Inspect index.html

```
PS C:\DockerLab\docker-web-demo> docker exec -it docker-web-container sh
/usr/share/nginx/html # pwd
/usr/share/nginx/html
/usr/share/nginx/html # ls
50x.html    index.html
/usr/share/nginx/html # cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>My Docker Web App</title>
</head>
<body>
  <h1>Welcome to Docker!</h1>
  <h2>My First Containerized Web Application</h2>

  <p>
    This web application is running inside a Docker container.
  </p>

  <p>
    Dockerfile → Image → Container
  </p>
</body>
</html>/usr/share/nginx/html # |
```

Container Lifecycle

Task 16: Exit Container

```
</html>/usr/share/nginx/html # exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug docker-web-container
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\DockerLab\docker-web-demo> |
```

Task 17: Stop Container

```
PS C:\DockerLab\docker-web-demo> docker stop docker-web-container
docker-web-container
PS C:\DockerLab\docker-web-demo> |
```

Task 18: View Stopped / All Containers

```
PS C:\DockerLab\docker-web-demo> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
adc9196ce9fb   mywebapp:v1   "/docker-entrypoint..." 8 minutes ago  Exited (0)   About a minute ago  docker-web-container
PS C:\DockerLab\docker-web-demo> |
```

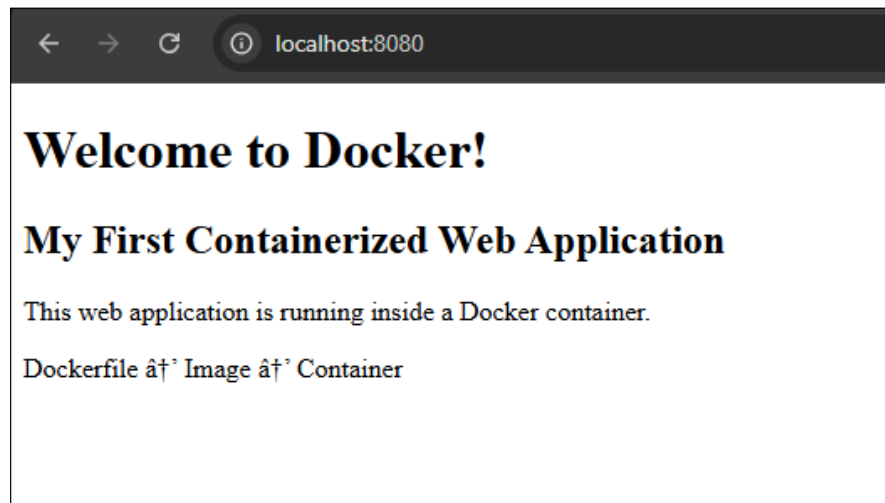
Task 19: Start Container

```
PS C:\DockerLab\docker-web-demo> docker start docker-web-container
docker-web-container
PS C:\DockerLab\docker-web-demo> |
```

Task 20: Verify Running Container

```
PS C:\DockerLab\docker-web-demo> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
adc9196ce9fb   mywebapp:v1   "/docker-entrypoint..." 8 minutes ago  Up 22 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  docker-web-container
PS C:\DockerLab\docker-web-demo> |
```

Task 21: Verify Web Page After Restart



Docker Hub Push

Task 22: Docker Hub Login

```
PS C:\DockerLab\docker-web-demo> docker login
Authenticating with existing credentials... [Username: vijayyh]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\DockerLab\docker-web-demo> |
```

Task 23: Tag Image

```
Login Succeeded
PS C:\DockerLab\docker-web-demo> docker tag mywebapp:v1 vijayyh/mywebapp:v1
PS C:\DockerLab\docker-web-demo> |
```

Task 24: Verify Tagged Image

```
PS C:\DockerLab\docker-web-demo> docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
mywebapp:v1	ed106ce1a492	93.6MB	26.3MB	U
vijayyh/mywebapp:v1	ed106ce1a492	93.6MB	26.3MB	U

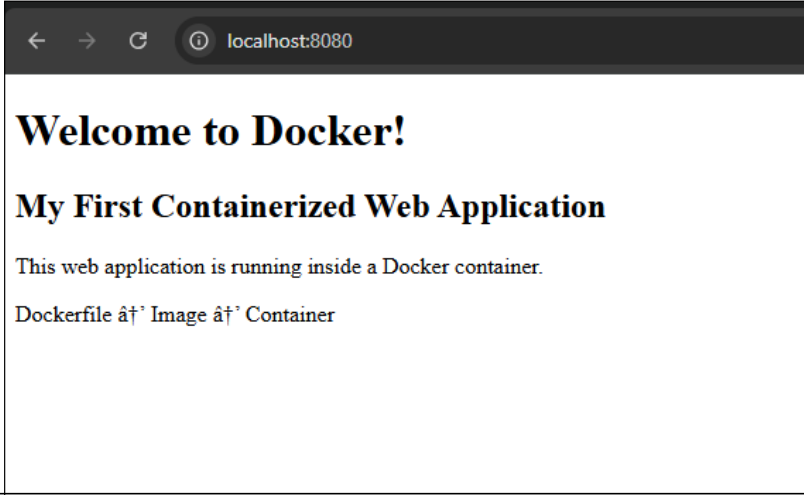
```
PS C:\DockerLab\docker-web-demo> |
```

Task 25: Push to Docker Hub

```
PS C:\DockerLab\docker-web-demo> docker push vijayyh/mywebapp:v1
The push refers to repository [docker.io/vijayyh/mywebapp]
2c43d02625c4: Pushed
44136fa355b3: Pushed
d9aae54b5831: Pushed
e2de96513ba9: Pushed
6c53d0b2a666: Pushed
745dfb2690dd: Pushed
9a9a644fdd6a: Pushed
64c8194480fe: Pushed
e76228b47809: Pushed
e72112c14215: Pushed
4f4fb700ef54: Pushed
f5e127468f9b: Pushed
76ebd212fae7: Pushed
v1: digest: sha256:ed106ce1a492811e6c20881a73f1534844e833c58bbb8cedccf8397e3db8babf size: 856
PS C:\DockerLab\docker-web-demo> |
```

Task 26: Verify Docker Hub Repository

The screenshot shows the Docker Hub web interface for the repository 'vijayyh/mywebapp'. The interface includes a sidebar with navigation options like 'Repositories', 'Collaborators', 'Settings', 'Billing', 'Usage', 'Public', and 'Storage'. The main content area displays the repository details, including a list of tags (v1) and a repository overview section. The repository overview section contains a description of the repository and a link to the repository overview page.

Docker Hub Pull & Final Verification	Task 27: Stop Container <pre>PS C:\DockerLab\docker-web-demo> docker stop docker-web-container docker-web-container PS C:\DockerLab\docker-web-demo> </pre> Task 28: Remove Container <pre>PS C:\DockerLab\docker-web-demo> docker rm docker-web-container docker-web-container PS C:\DockerLab\docker-web-demo> </pre> Task 29: Remove Local Docker Hub Tag <pre>PS C:\DockerLab\docker-web-demo> docker rmi vijayyh/mywebapp:v1 Untagged: vijayyh/mywebapp:v1 PS C:\DockerLab\docker-web-demo> </pre> Task 30: Pull Image from Docker Hub <pre>PS C:\DockerLab\docker-web-demo> docker pull vijayyh/mywebapp:v1 v1: Pulling from vijayyh/mywebapp Digest: sha256:ed106ce1a492811e6c20881a73f1534844e833c58bbb8cedccf8397e3db8babf Status: Downloaded newer image for vijayyh/mywebapp:v1 docker.io/vijayyh/mywebapp:v1 PS C:\DockerLab\docker-web-demo> </pre> Task 31: Run Pulled Image <pre>PS C:\DockerLab\docker-web-demo> docker run -d -p 8080:80 --name docker-web-container vijayyh/mywebapp:v1 bdd1e1665a65ded991976108af6a77799577cfff86def11a79b22857897dfff99e PS C:\DockerLab\docker-web-demo> </pre> Task 32: Final Browser Verification  The screenshot shows a web browser window with the address bar set to 'localhost:8080'. The page content includes a heading 'Welcome to Docker!', a sub-heading 'My First Containerized Web Application', and two paragraphs: 'This web application is running inside a Docker container.' and 'Dockerfile â†’ Image â†’ Container'.
Conclusion	The experiment successfully demonstrated Dockerfile-based containerization of a sample web application, image creation and execution, environment and file inspection, container stop/start operations, and the complete Docker Hub workflow of tagging, pushing, pulling, and rerunning the image.